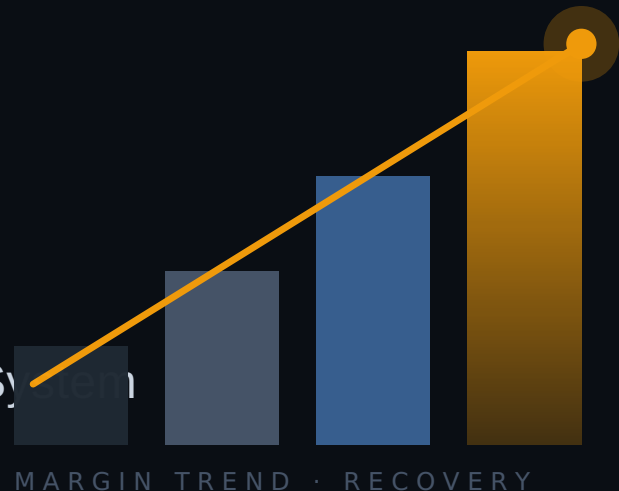

MarginOptimizer

B2B Dynamic Alternative Flight & Yield Management System

STRANDS AGENT AUTOMATED PNR ANALYSIS

CONFIDENTIAL · 2026



T A B L E O F C O N T E N T S

What This Document Covers

01 | The Cost Factor Extracting invisible profit from waiting tickets. **02 | The Solution Lifecycle** How the autonomous agent operates at 02:00 (Europe/Berlin). **03 | Infrastructure & Scaling** Processing thousands of items without triggering API bans (SQS). **04 | The Approval Loop (HITL)** Slack Webhooks and operator-side security. **05 | Development Roadmap** From local validation to cloud dashboard deployment.

THE COST FACTOR

Creating Money From Inefficiency

The Scenario: A package tour is sold today, departure in 3 months. The locked-in flight cost is €1,000. In month 2 a competitor drops an identical flight — same baggage, same window — to €800. Agencies miss it because no human can scan 10,000 PNRs every night.



ROI: ~50,000 yearly bookings × 2% profitable-hit rate × €40 average saving = **≥ €40,000 / year** in pure recovered margin.

SYSTEM ARCHITECTURE

AWS Serverless High-Throughput Engine



SERVICE	FUNCTION IN THE ARCHITECTURE
AWS EventBridge	Nightly cron at 02:00 Europe/Berlin kicking the ingest Lambda
Amazon SQS	Queues 10,000 bookings with a DLQ, shielding Traffics from bursts
Lambda (Worker)	One Strands + Haiku 4.5 agent per PNR, max 10 concurrent
API Gateway	Receives Slack approval webhooks, HMAC-verified before routing
Aurora Postgres	Writes every approved swap to <code>yield_events</code> ; feeds QuickSight

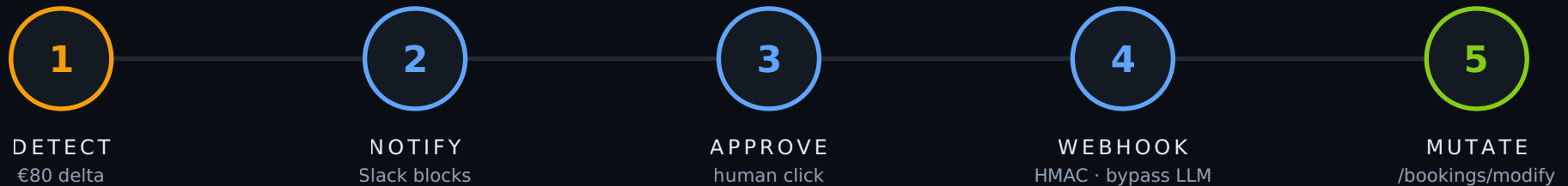
STRANDSTOLING

The Margin Finders

- `use_traffics` ▶ `/offers/{code}/alternativeFlights`
- Pings the Traffics spine to query identical alternative planes.
- `calculator_tool` ▶ deterministic (New – Old) delta
- Prevents LLM mathematical hallucinations — arithmetic is never model-generated.
- `slack_tool` ▶ operations channel `#yield-ops`
- Dispatches interactive UI blocks (Approve/Reject) showing the found margin.

THE APPROVAL LOOP

Zero Unwanted Actions



The agent runs fully autonomous up to step 2. At step 4 the Webhook **bypasses the LLM entirely** — deterministic code (and only deterministic code) ever touches production bookings. One locked path in, one locked path out.

INVESTMENT & ROADMAP

Execution Strategy



01 - Validation (0-1 mo) — Prove the prompt logic isolates profit perfectly, locally. **02 - Cloud Farm (1-2 mo)** — Deploy AWS SQS. Tune rate-limit handling for 429s. **03 - Operational Link (2-3 mo)** — Slack Webhooks + direct mutation scripts. **04 - Big Data Returns (4+ mo)** — Amazon QuickSight "Yield Dashboards" for C-level.

MarginOptimizer

Autonomous, Invisible, Highly Profitable

CONFIDENTIAL & PROPRIETARY B2B YIELD
OPTIMIZATION AI.